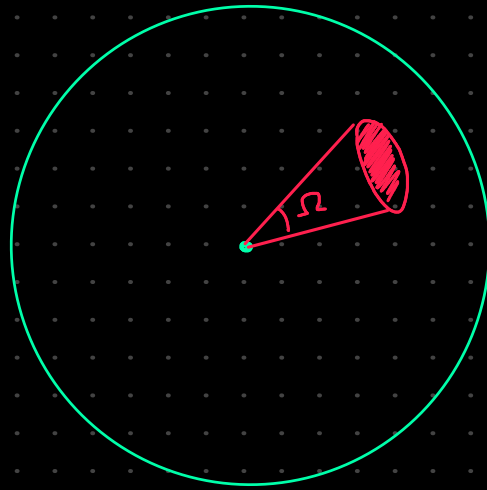
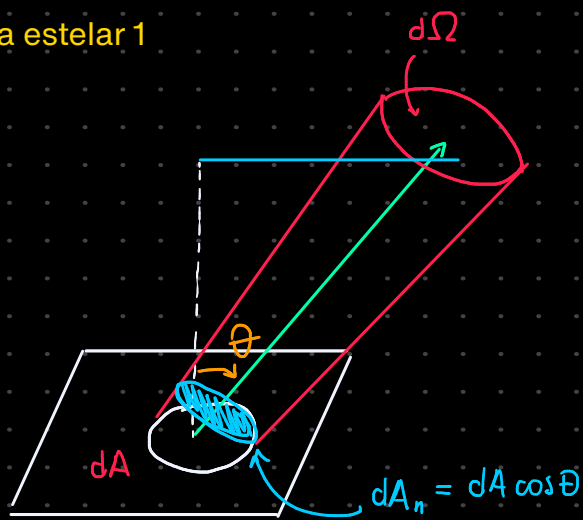


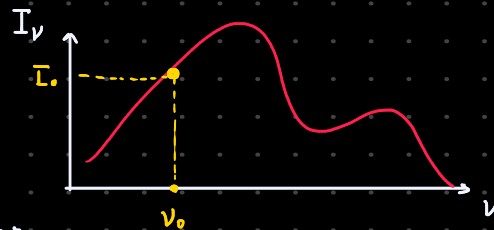


$$[\nu, \nu + d\nu]$$

$$dE_\nu = I_\nu \cos \theta dA d\nu d\Omega dt$$

↑ intensidad específica $[\text{W m}^{-2} \text{Hz}^{-1} \text{sr}^{-1}]$

$$I = \int_0^\infty I_\nu d\nu$$



densidad de flujo F F_ν $[\text{W m}^{-2}]$

unidad para el flujo jansky (Jy) $1 \text{ Jy} = 10^{-26} \text{ W m}^{-2} \text{Hz}^{-1}$

$$F_\nu = \frac{1}{dA d\nu dt} \int_S dE_\nu = \int_S I_\nu \cos \theta d\Omega$$

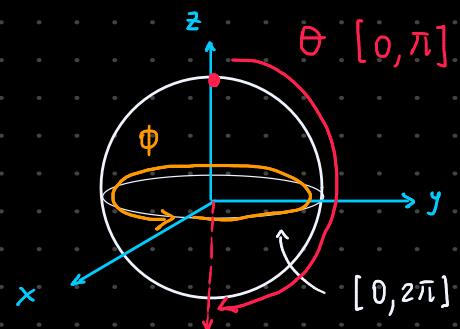
$$F = \int_S I \cos \theta d\Omega$$

↳ radiación isotrópica

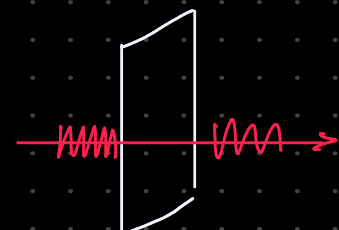
$$F = I \int_S \cos \theta d\Omega$$

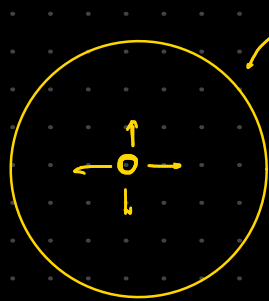
$$F = I \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \cos \theta \sin \theta d\theta d\phi$$

$$= I \int_0^{2\pi} d\phi \int_0^{\pi} \cos \theta \sin \theta d\theta = 0$$



$$d\Omega = \sin \theta d\theta d\phi$$

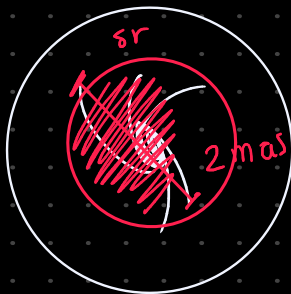




$$A = 4\pi r^2$$

$$L = 4\pi r^2 F$$

↖ $1/r^2$



brillo superficial

$$F \propto \frac{1}{r^2}$$

$$\Omega \propto \frac{1}{r^2}$$

constante

$$B = \frac{F}{\Omega} \propto \frac{\frac{1}{r^2}}{\frac{1}{r^2}}$$

2. Magnitudes

$$m = -2.5 \log_{10} \left(\frac{F}{F_0} \right)$$

↖ zero point

$$\begin{matrix} (F_m) \\ m \\ (F_{m+1}) \\ m+1 \end{matrix}$$

$$m - (m+1) = -2.5 \log_{10} \left(\frac{F_m}{F_0} \right) + 2.5 \log_{10} \left(\frac{F_{m+1}}{F_0} \right)$$

$$-1 = -2.5 \left[\log_{10} \left(\frac{F_m}{F_0} \right) - \log_{10} \left(\frac{F_{m+1}}{F_0} \right) \right]$$

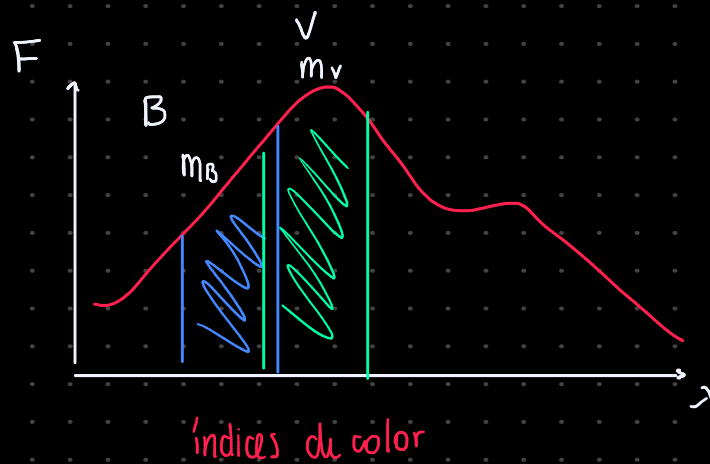
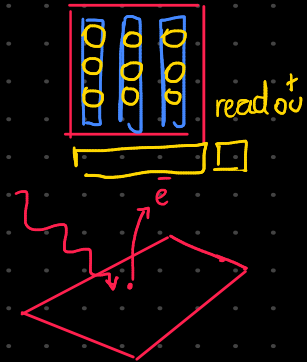
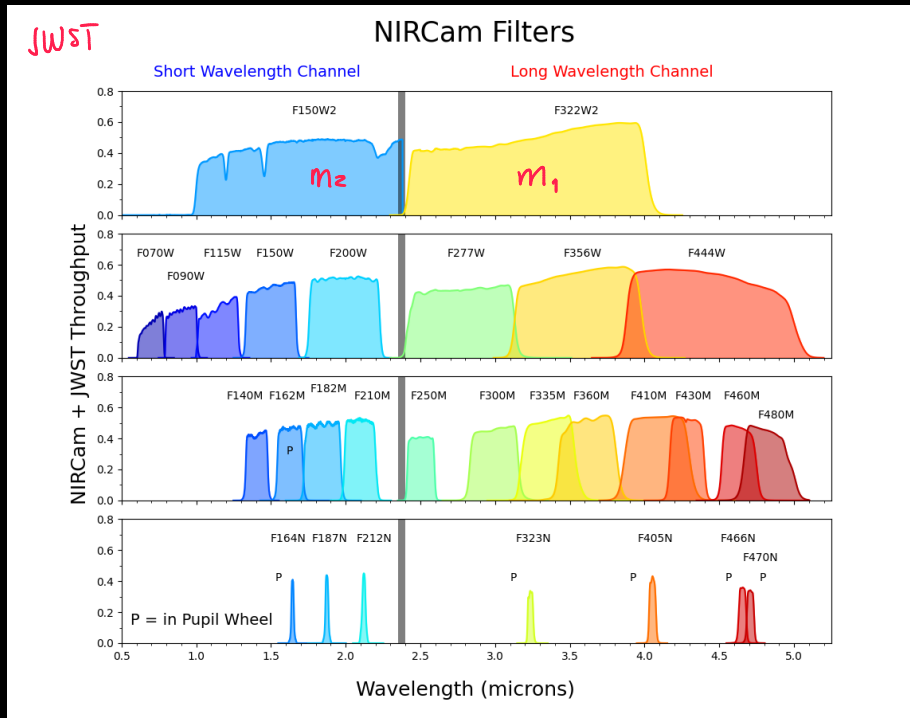
$$\frac{2}{5} = \log_{10} \left(\frac{F_m}{F_{m+1}} \right)$$

$$\frac{F_m}{F_{m+1}} = 10^{\frac{2}{5}} = \sqrt[5]{100} \quad \leftarrow \text{nuestra percepción}$$

$$\hookrightarrow m_1 - m_2 = -2.5 \log_{10} \left(\frac{F_1}{F_2} \right)$$

↪ magnitud bolométrica (m_{bol})

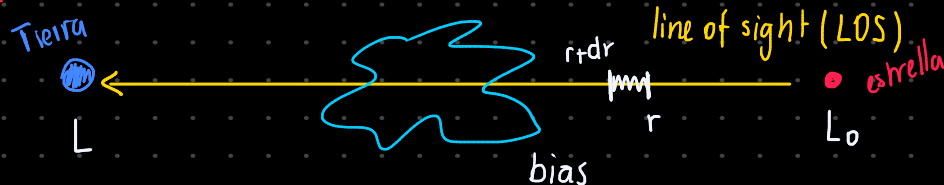
$$m_{bol} = m_v - BC$$



- zero point (AB system) $F_0 = 3631 \text{ Jy}$

↳ magnitud absoluta $m - M = 5 \log_{10} r - 5$
 ↑ en pc!

3. Extinción y optical thickness



pérdida de radiación $\rightarrow$ extinción (dL)

$[\nu, \nu+d\nu]$

$$dL = -\alpha L dr \quad (1)$$

$\uparrow$ opacidad $[\text{m}^{-1}]$

$\alpha = 0$ (vacío perfecto)

$\alpha = \infty$ (turbia)

optical thickness (τ)

$$d\tau = \alpha dr \quad (2)$$

(2) en (1)

$$dL = -L d\tau$$

$$\int_{L_0}^L \frac{dL}{L} = - \int_0^\tau d\tau$$

$$L = L_0 e^{-\tau}$$

$\uparrow$ disminuye exponencialmente

densidad de flujo en la superficie de la estrella F_0

en el instrumento $F(r)$

$$L = \Omega r^2 F(r)$$

$$L_0 = \Omega R^2 F_0$$

$$\hookrightarrow F(r) = F_0 \frac{R^2}{r^2} e^{-\tau}$$

$$F(10) = F_0 \frac{R^2}{(10)^2} e^{-\tau}$$

módulo de distancia

$$m - M = -2.5 \log_{10} \left(\frac{F(r)}{F(10)} \right)$$

$$= 5 \log_{10} \left(\frac{r}{10} \right) - 2.5 \log_{10} (e^{-\tau})$$

$$= 5 \log_{10} \left(\frac{r}{10} \right) + \tau (2.5 \log_{10} e)$$

$$m - M = 5 \log_{10} \left(\frac{r}{10} \right) + \underbrace{A}_{\text{mag}}, \quad A \geq 0$$

opacidad constante

$$\tau = \alpha \int_0^r dr$$

$$\tau = \alpha r$$

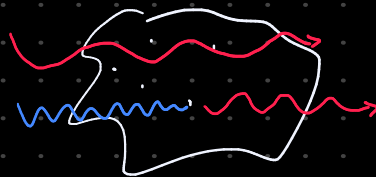
$$\rightarrow m - M = 5 \log_{10} \left(\frac{r}{10} \right) + \underbrace{\alpha r}_{\text{distancia}}$$

$$\alpha = 2.5 \alpha \log_{10} (e)$$

4. Colour excess

- reddening (enrojecimiento)

$B - V$
 negativo positivo



$$\left. \begin{aligned} V &= M_v + 5 \log_{10} \left(\frac{r}{10} \right) + A_v \\ B &= M_B + 5 \log_{10} \left(\frac{r}{10} \right) + A_B \end{aligned} \right\}$$

$$B - V = M_B - M_v + \underbrace{A_B - A_v}_{\text{color intrinsic}}$$

$$B - V = (B - V)_0 + E_{B-V}$$

color intrínseco

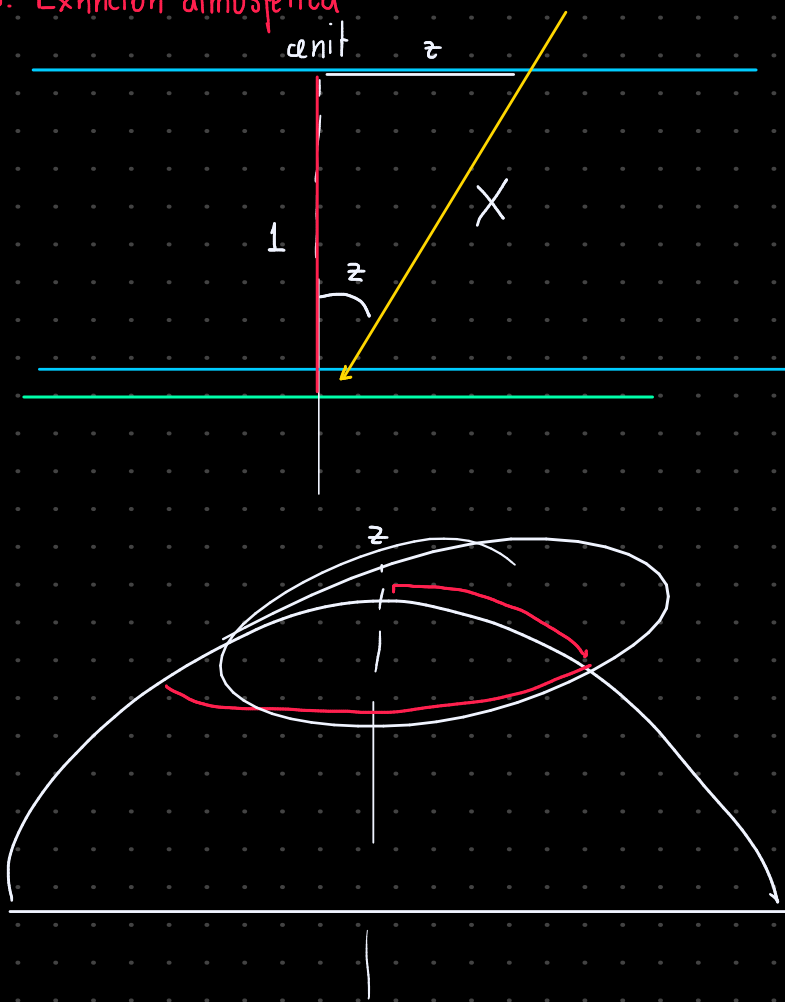
$$E(B - V) = E_{B-V} = A_B - A_v \quad \text{exceso de color}$$

$$R = \frac{A_v}{E_{B-V}} \approx 3.0$$

$$\rightarrow A_v \approx 3.0 E_{B-V}$$

$\rightarrow r$

5. Extinción atmosférica



m_0 (no atmósfera)
 z (d. cenital)

$$\cos(z) = \frac{1}{X}$$

$$X = \frac{1}{\cos(z)} = \sec(z)$$

air mass

$$m = m_0 + kX$$

coeficiente
de extinción

seeing

